Linear Pin Representations and the Limits of Patching Evidence in a Searchless Chess Transformer

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Abstract

An important question in mechanistic interpretability is whether neural networks trained on structured domains develop linear representations of abstract concepts, and whether intervening on these representations affects the model's output. We present a case study on the 270M searchless chess transformer of Ruoss et al. (2024), focusing on absolute pins. A linear probe identifies pin status well above bitboard, untrained-model, and random-label baselines. Mean-difference activation patching along the resulting pin direction reshapes the model's value-prediction output, but two controls reveal caveats to the behavioral interpretation. A placebo direction for the broader concept of geometric king–slider alignment produces a larger effect at matched magnitude despite being nearly orthogonal, and per-action scoring shows the model's recommended move rarely changes despite the much larger bucket-flip rate. Probe and patching peaks lie at different layers in both 270M and 9M variants of the architecture. Beyond our chess-specific findings, our results provide a quantitative case study of how standard interpretability evidence types, probe accuracy, random-direction controls, and output-bucket flips, can each overstate behavioral causation.

1. Introduction

The question of whether neural networks trained on structured domains learn structured internal representations has become one of the defining questions of mechanistic interpretability (Li et al., 2023; Nanda et al., 2023; Park et al., 2023). The *linear representation hypothesis* anticipates that advanced semantic concepts correspond to specific directions in activation space and are therefore detectable with linear probes. This prediction has been confirmed for board state in OthelloGPT (Nanda et al., 2023), human chess concepts in AlphaZero (McGrath et al., 2022), and for piece placement and player skill in small chess language models trained on PGN move sequences (Karvonen, 2024). The question we pursue is whether this hypothesis extends from concrete states (which piece sits on which square) to strategic ideas such as pins, forks, and king safety are dependent on the relationship between multiple pieces.

We specifically study this question for *absolute pins*, which are configurations in which a piece cannot move legally without exposing its king to check. Pins are a useful probing target. They are easy to detect from board state, frequent in middlegame play (base rate ≈19% in our sample), and, to our knowledge, have not been the subject of a targeted casual representation study in a chess model. The model itself is also unique. The 270M searchless transformer of Ruoss et al. (2024) is trained as a supervised regressor on Stockfish action-value targets, with no search at inference time and no self-play during training, and yet it achieves grandmaster-level Lichess Elo. To our knowledge there is no published work that considers the encoding of tactical structure in such a model, or whether any such encoding can be causally intervened upon to influence its output. Karvonen et al. (2024) reports a sparse-autoencoder feature labeled "pin" on a smaller PGN-only model (Chess-GPT), but with F1 of only 0.20 and does not validate causality. Likewise, the recent layer-wise alignment analysis of the Ruoss model by Lomasov et al. (2025) differs in that it focuses on positional ideas like center control instead of tactical motifs.

We make three empirical contributions. First, pins are linearly decodable in the 270M residual stream, peaking at AUC 0.919 at layer 3, above bitboard (+0.145 AUC) and untrained-model (+0.143 AUC) baselines. Second, mean-difference patching along the layer-6 pin direction yields a monotonic, direction-specific return-bucket effect (36.8% flip rate, 6.2× random; strictly monotonic in α), but placebo and per-action controls sharply limit the behavioral interpretation: a line-attack direction produces more bucket flips than pin in 96% of bootstrap resamples, and recommended moves change on only 2.8% of positions (1.73× random, $p = 0.054$). Third, probe and patching peaks are separated across both 270M (3 layers) and 9M (2 layers) models, and

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cross-layer norm-controlled patching supports a distinct-layer-encoding interpretation. Taken together, these results indicate that probe accuracy, random-direction specificity, and bucket flips are all more limited than would be expected under a strong behavioral-causality interpretation. Code, probe weights and patch directions will be released upon acceptance.

2. Background

The searchless chess transformer. Ruoss et al. (2024) train decoder-only transformers as supervised regressors on (s, a, q) tuples where s is a board position, a is a candidate UCI move, and q is a Stockfish-derived win-probability target, discretized into 128 buckets. The model receives a 77-token FEN encoding followed by a 2-token UCI action (79 tokens total) and predicts a log-softmax over return-value buckets. The 270M variant has 16 transformer blocks, a 1024-dimensional residual stream, 8 attention heads, SwiGLU MLPs with 4× widening, and pre-LayerNorm with a final post-LN; it is trained on Stockfish-annotated Lichess games without search or self-play. A 9M variant uses the same architecture with 8 transformer blocks, a 256-dim residual stream, 8 attention heads, and the same 4× SwiGLU widening. Both checkpoints are publicly available under CC-BY 4.0.

Absolute pins. A piece is *absolutely pinned* when moving it would place its own king in check, so the move is illegal. Absolute pins are distinct from *relative* pins (the pinned piece may legally move, but doing so loses material) and from *skewers* (the more valuable piece is in front of the less valuable one). We choose to study absolute pins because `python-chess`'s `board.is_pinned` gives a definite binary answer for any position. An example of an absolute pin is illustrated in Figure 1: with White to move, a Black bishop on a7 pins a White knight on d4 against the White king on g1. Therefore, the knight cannot legally move.

Labeling convention (either-side). The label we use, `has_pin` is true if *any* piece on the board is absolutely pinned, regardless of which side's king would be exposed by moving the pinned piece. To further clarify, we end up marking pins for both the side-to-move and their opponent. The most behaviorally relevant pins are pins of the side-to-move's pieces, since these are the pins that constrain the actual legal moveset, but pins of their opponent's pieces are strategically important (they create tactical opportunities for the side-to-move), so we include them in our labels. We use this convention consistently for probe training, paired-positions construction, the line-attack placebo concept (Section 4), and all reported results. A side-to-move-only analysis is a possible follow-up.

Probing and interventions. *Linear probes* (Alain & Bengio, 2017; Belinkov, 2022) train a logistic regression on activations to predict a concept label, providing correlational evidence that the concept is linearly decodable. Hewitt & Liang (2019) caution that probe accuracy can reflect probe capacity rather than the actual model structure. They suggest instead using *selectivity*, the gap between true-label accuracy and accuracy on a control task with random labels, as a more conservative measure. *Activation patching* (Vig et al., 2020; Meng et al., 2022; Zhang & Nanda, 2024) provides causal evidence by intervening on intermediate activations and measuring how the final output shifts.

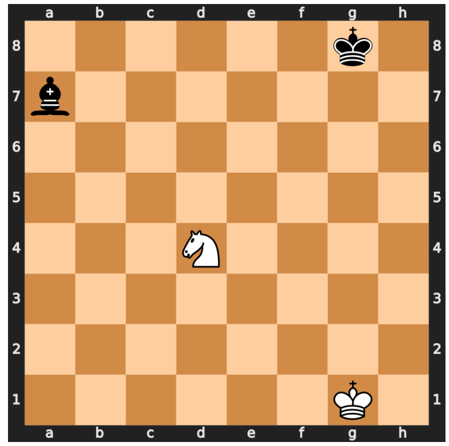

Figure 1. A canonical absolute pin (White to move). The Black bishop on a7 pins the White knight on d4 against the White king on g1. Any knight move would expose the king to check, so the knight has no legal moves.

3. Probing for Pin Representations

3.1. Dataset

We first sample positions from the Lichess open database (February 2026), filtering to games where both players are rated $\geq$1800 Elo and the time control is rapid or classical (base time $\geq$180s). For each qualifying game, we sample two random positions from plies 20–100. From 100,000 raw positions, we initially observe a pin base rate of 19.16%.

To avoid confounding variables, such as pins occurring with material imbalance or during specific game phases, we construct a balanced dataset differentiated by both *material count* (4 buckets) and *move number* (4 buckets). Within each 4×4 cell we sample equal numbers of pin-present and pin-absent positions. This yields 38,312 positions total (19,156 pin-present; 19,156 pin-absent), split 70/15/15 into train/val/test by hashing on `game_id` so that no two positions from the same game appear in different splits. Because the hash split groups by game rather than by position, per-split pin rates are only approximately equal (49.28% val, 49.99% test) rather than exactly 50%. These small deviations from 50% do not materially affect AUC, and the balanced construction prevents other sources of imbalance from explaining the reported probe performance.

3.2. Probe architecture and controls

For each layer $\ell \in \{0, 1, \ldots, L\}$ ($\ell = 0$ is the post-embedding residual stream, $\ell = L$ after the final transformer block) we cache mean-pooled residual-stream activations $\mathbf{h}_\ell \in \mathbb{R}^d$ over all 79 token positions. Mean-pooling allows us to avoid encouraging a particular token position. We train an L2-regularized logistic regression on the train split with C selected on val (scikit-learn `LogisticRegression`, L-BFGS, $C \in \{0.01, 0.1, 1.0, 10.0\}$).

Additionally, we include four controls: (i) a *random-label baseline* (shuffled pin labels within the train split. The gap between true and shuffled AUC is the *selectivity*, inspired by (Hewitt & Liang, 2019)); (ii) bootstrap 95% CIs from 1000 resamples of the test set; (iii) *cross-architecture replication* on the 9M variant; (iv) a *bitboard baseline* that trains the same probes on raw piece-on-square indicator features, bypassing the transformer entirely to establish how much pin information is recoverable from the board encoding alone. We additionally report MLP-probe and untrained-model controls in Appendix C.

3.3. Probe results

The 270M probe peaks at layer 3 with AUC 0.919 (95% bootstrap CI [0.912, 0.925]) and selectivity 0.328, then declines steadily to AUC 0.759 by layer 16. Random-label probes remain near chance (0.50) at all layers, supporting the interpretation that probes are reading label-relevant structure. Figure 2 shows the full trajectory and the layer-wise table is in Appendix A. The pattern vaguely resembles the classic inverted U. It demonstrates that pin information becomes rapidly more linearly decodable in the early layers (1–3), holds for two further layers, then declines, possibly because these features are integrated with other information for downstream move-value computation. We replicate this pattern on the 9M model (Table 7, Appendix B), where the peak occurs later (layer 5 of 8) and is weaker in absolute terms (peak AUC 0.825, selectivity 0.253). The earlier and stronger peak in the 270M model is suggestive of the broader trend that larger models may form abstract features earlier in their depth-normalized layer index (Templeton et al., 2024). Despite this, we caution that two model sizes from one architecture family is weak evidence of such a trend, and this comparison should not be read as confirming.

Bitboard baseline. A natural concern is that the high probe AUC reflects information already accessible from raw piece-on-square geometry, which the trained transformer would simply preserve rather than learn. In order to address these concerns, we train the same probes (linear and MLP) on

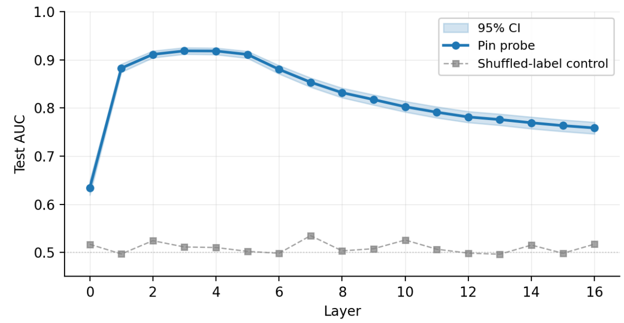

Figure 2. Layer-wise pin probe AUC on the 270M model (blue) versus shuffled-label control (gray). Shaded region: 95% bootstrap CI. Peak at layer 3 (AUC 0.919, selectivity 0.328). The control sits at chance across all layers, supporting the interpretation that the probe reads label-relevant structure rather than fitting shuffled labels.

raw bitboard features, the 64 squares $\times$ 12 piece-color combinations plus side-to-move, yielding 769 binary features and bypassing the transformer entirely. The bitboard linear probe achieves test AUC 0.774 ([0.763, 0.786]); the bitboard MLP probe achieves 0.830 ([0.819, 0.840]). Both are substantially below the trained-model probe AUCs at layer 3 (0.919 linear, 0.926 MLP), as the trained 270M model improves over the bitboard linear baseline by approximately 0.145 AUC. The bitboard linear AUC is also approximately equal to the untrained-model peak (0.776 at layer 1; Appendix C), consistent with the interpretation that the untrained architecture and tokenization already make some board-geometry information linearly accessible, while training adds an additional recoverable component.

3.4. Where pins are represented

Two features of the layer-wise curve are worth flagging. First, the rise from layer 0 to layer 1 is exceptionally sharp: AUC jumps by 0.249 in a single block. The post-embedding representation already carries some pin-relevant signal (AUC 0.634), presumably because the FEN tokenization makes piece-on-square information directly accessible. It also indicates that extracting the relational structure of a pin, which depends on attacker, pinned piece, and king all lying on the same line, requires additional computation beyond the embedding layer. Most of the probe gain appears after a single transformer block.

Second, the post-peak decline is gradual rather than sudden. Between layers 3 and 16, AUC drops by 0.16 and selectivity by 0.15. Pin information is not completely erased, but its linear decodability declines, consistent with the possibility of integration into other features. This matches the pattern McGrath et al. (2022) report for AlphaZero concepts and Karvonen (2024) report for Chess-GPT board state, and motivates the causal analysis in the next section, as "where pin information is most decodable" is not the same question as "where intervention on that information most affects the model's output."

Linear vs. nonlinear accessibility. A linear probe with peak AUC 0.919 establishes that pin information is linearly separable in the residual stream, but does not bound how much information is available in principle. We trained MLP probes (single hidden layer, sizes $\{64, 128, 256\}$ selected on validation AUC, ℓ_2 weight decay 10^{-3}) on the same activations at each model's peak layer. The 270M MLP probe achieves test AUC 0.926 ([0.920, 0.932]) at layer 3, exceeding the linear probe by only 0.007 AUC; the 9M MLP probe achieves 0.847 ([0.837, 0.856]) at layer 5, a gap of 0.022 AUC. The negligible gap at the 270M scale indicates the shallow MLP recovers little additional pin information beyond the linear probe in this representation. The slightly larger gap in the smaller model is consistent with the observation that representational geometry becomes more effective with scale (Templeton et al., 2024). Full results in Appendix C. As such, we proceed with the linear-direction interpretation.

4. Activation Patching

Probing establishes that pin information is linearly accessible in the residual stream, yet it does not establish that the pin direction is functionally connected to the model's output (Belinkov, 2022). To test for such a connection, we follow the design of Karvonen (2024) and Nanda et al. (2023) and intervene on intermediate activations.

4.1. Method

Paired positions. We construct paired clean/corrupted inputs by scanning Lichess games for consecutive plies in which the either-side `has_pin` label changes (a pin appears or disappears across one move). This yields 2,000 paired positions $(s_{\text{pin}}, s_{\text{no-pin}})$ from 368 unique games. The pairs are tightly matched on confounding variables. Mean material count is 21.1 (pin) vs. 21.0 (no-pin) and mean material difference is 0.34 pieces. Pairs span move-numbers 11–204, ranging from the opening to the very late endgame. We hold out 500 pairs as a test set and use the remaining 1,500 to estimate the patching direction.

Mean-difference patching. For each layer ℓ, we compute

$$\Delta \mathbf{h}_\ell = \frac{1}{N} \sum_{i=1}^{N} \left(\mathbf{h}_\ell(s_{\text{pin}}^{(i)}) - \mathbf{h}_\ell(s_{\text{no-pin}}^{(i)}) \right), \qquad (1)$$

and at test time, given a position with $\texttt{has\_pin} = 1$, run the model with activations unmodified up to layer ℓ, then add $-\Delta \mathbf{h}_\ell$ to the layer-ℓ residual stream and continue the forward pass. This pushes the representation toward the no-pin manifold. Mean-difference patching is purposefully conservative, as it does not condition on the specific corrupted example (avoiding example-specific leakage) but is weaker than swap-style patching.

Output and behavioral metrics. The 270M model is an action-value predictor: for a tokenized (state, action) input it outputs a 128-dimensional distribution over discretized return-value buckets (Ruoss et al., 2024). For the bucket-flip experiments we use a fixed placeholder action token (index 0, `a1b1`); because this placeholder is not necessarily legal, we interpret bucket-flip results only as state-conditioned value-output perturbations, not as behaviorally meaningful action evaluations. The per-action experiment (Section 4.2) instead fills the action slot with each legal move and scores expected returns. We measure three quantities on the 500 held-out test positions: (i) *KL divergence* between patched and unpatched return-bucket distributions; (ii) *return-bucket flip rate*, the fraction of positions where the argmax bucket changes; and (iii) *logit difference* between the maximum logit of patched and unpatched bucket distributions.

4.2. Results

Layer-wise patching on the 270M model peaks at layer 6: the return-bucket flip rate reaches 36.8% (Wilson 95% CI [0.327, 0.411]), compared with 14.0% at layer 0 and 7.8% at layer 16. The flip rate exhibits a clear inverted-U pattern, with the middle layers (5–7) combining moderate KL with concentrated flips in the argmax bucket (consistent with a more targeted bucket-level effect rather than broad distributional perturbation). Logit differences remain near zero at the patching peak (≈ 0.002 at layers 5–7), indicating the intervention changes the top return bucket without producing a large uniform value shift. The full layer-wise table is in Appendix A.

The Intervention To understand intervention, it is important to clarify what flip rate measures. The model's output at the final token position is a 128-bucket distribution over expected return. Our flip rate counts positions where the highest-probability return bucket changes between the unpatched and patched forward passes. It is therefore a measure of how strongly the intervention reshapes the model's value-prediction at this layer, not a direct measure of which move the model would recommend. The full move-value computation in the searchless action-value design requires scoring each candidate action against the position separately (Ruoss et al., 2024). Here, the bucket-flip analysis uses the fixed placeholder action, while the per-action experiment below performs the full scoring loop. We read the combination of high return-bucket-flip rate plus near-zero logit-diff as evidence that the intervention shifts how the model represents value internally without changing what value it predicts. The patched distribution concentrates probability on a different bucket, but one that corresponds to roughly the exact same

Table 1. Random-direction control on the 270M model. $K = 10$ random unit directions of matched norm tested at each patching-effect plateau layer on the same 500 paired positions used for the main results. Pin-direction flip rate exceeds the random-direction maximum at every layer.

Layer	Pin flip rate	Random mean $\pm$ std	Random max
5	0.294	0.055 ± 0.007	0.066
6	**0.368**	0.059 ± 0.014	0.080
7	0.322	0.059 ± 0.009	0.070

expected return. This possibly results from the intervention leading to a value-relevant pin/king geometry feature rather than disrupting the output.

Directional specificity. To test whether the bucket-level effect is specific to the pin direction rather than generic perturbation sensitivity, we compared pin-direction patching to matched-norm random-direction patching at each of layers 5–7. For each layer we sampled $K = 10$ random unit directions, scaled each to $\|\Delta\mathbf{h}_\ell\|$, and ran the same patching pipeline. Random-direction flip rates concentrate tightly around 5–8% (layer 6: 0.059 ± 0.014, max 0.080). The pin-direction flip rate at layer 6 is 0.368, approximately $6\times$ the random mean and $4.5\times$ the random maximum. The pattern holds at adjacent layers (Table 1). This shows the bucket-level effect is not a generic response to large residual-stream perturbations. Pin-direction patching produces much larger effects than random directions of equivalent magnitude. We return in a later section to whether this directional specificity holds against meaningful (non-random) alternative directions.

Dose–response sweep. A second check is whether the output effect scales monotonically with the magnitude of our interventions. We swept the patching coefficient $\alpha \in \{0, 0.5, 1, 2, 5\}$ at layer 6, patching with $\alpha \cdot \Delta\mathbf{h}_6$ on the same 500 held-out pairs. The flip rate is strictly monotonic in α (0.000, 0.200, 0.368, 0.564, 0.674) and saturates above $\alpha = 2$. Mean KL grows roughly quadratically (0.000, 0.011, 0.042, 0.157, 0.623). We caution that monotonicity is not by itself concrete evidence of *concept-specific* causal structure. Many directions (including completely random ones) likely produce monotonic output disruption as norm increases, and we did not run matched dose–response curves for random or placebo directions.

Placebo-concept control. The random-direction baseline rules out generic perturbation sensitivity, but it is also necessary to ask whether the bucket-level effect is specific to the *pin* concept or to a broader concept of which pin is an instance. A placebo direction trained on the broader "line-attack" concept produces a larger flip rate than pin at matched magnitude (0.557 ± 0.099 across $B = 50$ bootstrap resamples vs. pin's 0.368, exceeding pin in 48/50 resamples; Table 2) despite being nearly orthogonal to it (cosine 0.076 ± 0.030). We describe the construction in the following information.

Table 2. Placebo-concept comparison on the 270M model. All directions are computed and applied at layer 6, evaluated on the same 500 held-out pin-paired test positions; flip rate is over the highest-probability return-value bucket. The bootstrap row reports mean $\pm$ std across $B = 50$ resamples of 200 line-attack pairs each.

Direction	Norm	Flip rate
$\Delta\mathbf{h}_{\text{pin}}$ (native)	1094	0.368
$\Delta\mathbf{h}_{\text{line}}$ (native, single sample)	1444	0.688
$\Delta\mathbf{h}_{\text{line}}$ (rescaled, single)	1094	0.632
$\Delta\mathbf{h}_{\text{line}}$ (rescaled, bootstrap)	1094	$\mathbf{0.557 \pm 0.099}$
bootstrap 95% percentile range		$[0.368, 0.754]$

We define line-attack as geometric alignment, ignoring intervening pieces: either side's king lies on the same rank, file, or diagonal as an enemy rook, bishop, or queen whose movement type is compatible with that ray (rook on rank or file, bishop on diagonal, queen on either), with no requirement that the line be unobstructed. Under this definition every absolute pin is also a line–attack, but many line-attack positions are not pins. We labeled all 38,312 positions in our balanced dataset ($P(\text{line-attack}) = 0.644$, with $P(\text{line-attack} \mid \text{pin}) = 1.000$), constructed 250 line-attack-present/line-attack-absent paired training examples using the same consecutive-position matching procedure used for pins. The single-sample $\Delta\mathbf{h}_{\text{line}}$ uses the first 200 pairs, the bootstrap uses $B = 50$ resamples of 200 pairs each (with replacement) drawn from the full pool of 250, on the same 500 pin-paired test positions. Although this uses fewer pairs than the pin direction (1,500), the bootstrap resampling directly measures the stability of the resulting direction and flip rate.

The line-vs-pin gap complicates the "directional specificity" reading of our patching results. The pin direction has a reproducible output effect and is nearly orthogonal to a related concept's direction, but at matched magnitude the placebo produces a *larger* bucket-level effect along an essentially different vector in the great majority (96%) of bootstrap resamples. We read this as evidence that (i) the pin direction encodes pin-related information distinct from line-attack-related information (high and stable orthogonality), but (ii) "directional specificity" established against random baselines should not be read as "the pin direction is the unique or even the dominant concept-encoding direction in this region of residual stream." Multiple king-geometry concepts each appear to have linear directions with substantial bucket-level effect on the value-prediction output, and probing for a single concept in isolation cannot establish which one drives any given intervention. We discuss the implications of this finding in Section 6.

Per-action scoring: from bucket flips to move recommendations. The 36.8% flip rate measures change in the model's highest-probability return-value bucket for a fixed (state, action) input where the action is a placeholder. This is not the same as change in the model's *recommended move*, which requires scoring all legal moves from each position (the standard inference procedure for the searchless action-value model; Ruoss et al. 2024). We therefore ran the full per-action scoring loop on the same 500 held-out test positions. For each position, we enumerate legal moves, build a batched input with each move encoded into the action slot, run both clean and patched (layer-6 pin-direction) forward passes, and compare the argmax over expected returns under each.

The model's recommended move changes on only **14 of 500 positions (2.8%, Wilson 95% CI [1.7%, 4.6%])** after pin-direction patching at layer 6. This is striking given the 36.8% return-bucket flip rate on the same positions: *the intervention substantially reshapes the value distribution but rarely changes which move the model would actually recommend.* We interpret this as evidence that pin-direction patching can shift value estimates without usually changing the top legal move. The win-probability difference between clean and patched recommendations at the few positions where the move does change is small (median 0.009, max 0.012), consistent with the return-bucket-flip-but-near-zero-logit-diff pattern reported in Table 6.

To anchor the 2.8% rate against random perturbation, we ran the same per-action scoring procedure with $K = 10$ matched-norm random unit directions at layer 6 (Table 3). Random directions change the recommended move on $1.62\% \pm 0.21\%$ of positions (mean $\pm$ std across $K = 10$ directions, maximum 2.00%). The pin rate of 2.80% exceeds the empirical random-direction maximum and is $\sim$1.73$\times$ the mean random rate. As a heuristic, a pooled two-proportion z-test comparing pin (14/500) against pooled random (81/5,000) gives $z = 1.93$, $p = 0.054$ (two-sided), placing the difference near the conventional 0.05 threshold. We note this p-value should be interpreted cautiously because the random-direction trials reuse the same 500 test positions across directions, violating the independence assumption a strict pooled z-test requires. We interpret pin as plausibly but not conclusively distinguishable from random at the move-recommendation level. This $1.73\times$ multiplier is much smaller than the $6.2\times$ specificity at the return-bucket level (0.368 vs. 0.059). The ratio of these multipliers (roughly $3.6\times$ shrinkage from bucket to move) is a quantitative measurement of how much the directional-specificity finding attenuates as the metric moves closer to model behavior.

Table 3. Per-action scoring directional-specificity comparison on the 270M model at layer 6, with Wilson 95% confidence intervals. The pin direction is descriptively higher than the sampled random directions at the move-recommendation level but with a much smaller multiplier than at the return-bucket level, and pin-vs-random at the move level is near the conventional 0.05 threshold ($p = 0.054$ two-sided). The random Wilson CI pools direction-position evaluations and is descriptive: random directions share the same underlying 500 positions.

Metric	Pin	Random (mean $\pm$ std)
Return-bucket flip rate	0.368	0.059 ± 0.014 (K=10)
Recommended-move change rate	0.028	0.0162 ± 0.0021 (K=10)
Wilson 95% CI	$[0.017, 0.046]$	$[0.013, 0.020]$ pooled
Pin-to-random ratio (bucket)	6.2$\times$	—
Pin-to-random ratio (move)	1.73$\times$	—

Table 4. Cross-layer patching on the 270M model. Each row patches a mean-difference direction (column *Direction*) at a target layer (column *Patch layer*), preserving the direction's native ℓ_2 norm.

Direction	Norm	Patch layer	Flip rate
$\Delta\mathbf{h}_3$	144	3 (own)	0.156
$\Delta\mathbf{h}_6$	1094	6 (own)	**0.368**
$\Delta\mathbf{h}_3$	144	6 (cross)	0.014
$\Delta\mathbf{h}_6$	1094	3 (cross)	0.702

4.3. The probe–patching gap

Comparing Table 5 and Table 6, the probe peak (layer 3) and the patching peak (layer 6) lie three layers apart. Figure 3 overlays the two curves, showing that pin information is most decodable in the early layers (ℓ 1–3), remains decodable across layers 4–6, and gradually loses linear decodability in later layers. The patching effect on the value-prediction output is largest precisely where the probe accuracy is already past its peak but still high.

Are the layer-3 and layer-6 directions the same vector? A natural first reading of the gap is that the pin feature is *written* into the residual stream at layer 3 (where probes peak) and *read* downstream at layer 6 (where intervention has the largest output effect), with the same vector being used at both layers. To probe this, we ran cross-layer patching: applying $\Delta\mathbf{h}_3$ at layer 6, and applying $\Delta\mathbf{h}_6$ at layer 3, on the same 500 held-out pairs.

The results (Table 4) show a striking asymmetry. $\Delta\mathbf{h}_3$ patched at layer 6 produces only 1.4% flips (below the 5–8% random-direction baseline), while $\Delta\mathbf{h}_6$ patched at layer 3 produces 70.2%. The layer-3 direction at its own layer produces 15.6% flips, less than half of $\Delta\mathbf{h}_6$'s 36.8%. Combined with the fact that the layer-3 probe is more accurate (AUC 0.919), this suggests the layer-3 encoding is the more *decodable* but less *output-consequential* of the two. The two directions also have very different magnitudes: $\|\Delta\mathbf{h}_6\|/\|\Delta\mathbf{h}_3\| \approx 7.6$. Two readings of the asymmetry are

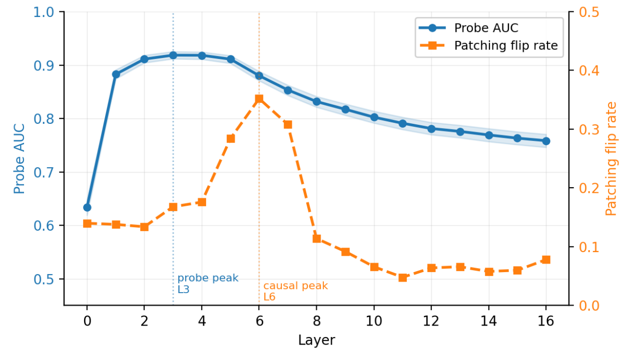

Figure 3. Probe AUC (blue) and patching flip rate (orange) on the 270M model, plotted against layer. Peaks differ by three layers: probe peaks at layer 3 (AUC 0.919) while patching effect peaks at layer 6 (flip rate 0.368). Shaded regions: 95% CIs.

possible: either the directions are similar but $\Delta\mathbf{h}_3$ is too small to matter, or they are genuinely different vectors with the magnitude difference partially explaining the patching asymmetry.

Norm-controlled follow-up. We rescale each direction to the other layer's norm and re-run (an out-of-distribution intervention because residual-stream activation magnitudes differ by layer, but one that constrains the same-vector vs. distinct-vector readings). With $\Delta\mathbf{h}_3$ rescaled 7.6× and patched at layer 6, the flip rate is 0.100. $\Delta\mathbf{h}_6$ rescaled *down* and patched at layer 3 produces 0.096. The 70% effect was therefore largely magnitude-driven, but the 10% remaining for matched-magnitude cross-layer cases is above the 5–8% random-direction rates for native layer-6 patching. We read this as consistent with distinct vectors that share some functional subspace: the model appears to encode pins linearly at multiple processing stages, with the layer-6 encoding more directly connected to the value-prediction output, but the two encodings are not identical or pure rescalings of each other.

This refines the methodological lesson: the probe–patching gap here does not appear to reflect a single feature being written and read at different layers, but rather distinct layer-specific encodings of the same concept. Further mechanistic decomposition (e.g., distributed alignment search or sparse-autoencoder analysis at the relevant layers) would clarify how these encodings relate, and we leave this to future work.

The 9M model shows the same qualitative pattern with a smaller offset: probe peak at layer 5 (AUC 0.825), patching peak at layer 7 (flip rate 0.306), giving a 2-layer gap (Appendix B). The fact that this gap appears in both models we study and persists under matched-pair design suggests it is consistent across the two scales of this architecture, although we caution that two model sizes from one family is not strong evidence of generality beyond the searchless transformer family. The dose–response sweep also replicates cleanly on the 9M model: flip rate is strictly monotonic in α (0.000, 0.162, 0.316, 0.490, 0.752 at $\alpha \in \{0, 0.5, 1, 2, 5\}$). The cross-layer + norm-controlled experiment partially replicates: with $\Delta\mathbf{h}_5$ rescaled to $\|\Delta\mathbf{h}_7\|$ and patched at layer 7, the 9M flip rate is 0.232, against 0.316 for $\Delta\mathbf{h}_7$ at its own layer. The 9M norm-matched cross-layer gap (0.232 vs. 0.316) is smaller than the 270M analogue (0.100 vs. 0.368), suggesting that the layer-specific specialization of pin encodings is more pronounced in the larger model, though we again flag two model sizes as weak evidence for any scale-related claim. Full 9M tables are in Appendix B.

This finding has methodological implications. Hewitt & Liang (2019) warned that probe accuracy can be a misleading proxy for the causal role of a feature. Our results provide a concrete, quantitative example: had we relied on probe accuracy alone, we would have located the pin representation at layer 3, but the layer where intervening on that representation most reliably perturbs the model's value-prediction output is three layers downstream.

5. Related Work

Prior work has shown that game models learn interpretable internal structure: AlphaZero encodes human chess concepts (McGrath et al., 2022), OthelloGPT learns board state (Li et al., 2023; Nanda et al., 2023), and Chess-GPT represents piece placement and player skill (Karvonen, 2024), with sparse-autoencoder work reporting a weak "pin" feature (Karvonen et al., 2024). Other work studies learned look-ahead in Leela (Jenner et al., 2024; Cruz, 2025) and concept alignment in the Ruoss model (Lomasov et al., 2025). Our work differs by targeting a tactical motif in the Ruoss searchless action-value architecture and by pairing linear probes with activation patching, placebo-concept controls, per-action scoring, and cross-layer analyses. Methodologically, we build on probing/selectivity (Alain & Bengio, 2017; Hewitt & Liang, 2019; Belinkov, 2022) and activation patching (Vig et al., 2020; Meng et al., 2022; Zhang & Nanda, 2024), while leaving SAE/DAS decompositions (Geiger et al., 2024; Bricken et al., 2023) to future work.

6. Discussion and Limitations

6.1. Scope and intervention strength

Single-model scope. Our results characterize one model (270M searchless transformer) plus a 9M replication. We therefore can not generalize, and can not claim that all chess transformers represent pins linearly. We believe the Ruoss model is particularly worth mapping because its training signal (Stockfish action-values, no search, no self-play) is unlike that of AlphaZero, Chess-GPT, or Leela, so structural findings on it bear on the question of how training signal shapes internal representation.

Conservative intervention strength. Our 36.8% return-bucket flip rate is moderate, not overwhelming. Two factors contribute. First, mean-difference patching estimates a single direction and applies it uniformly, ignoring the per-example structure that swap-style patching exploits. More expressive alignment methods (Geiger et al., 2024) or per-example optimal-transport matching may yield larger output effects, but they would also need the same placebo and move-level controls we report here. Second, many positions have a clearly best return-value bucket, and a flip on argmax requires the patched distribution to cross a winner-take-all threshold over 128 buckets. In contrast, KL divergence (which does not have this discreteness) shows clean layer-wise structure even where flip rate is small.

What the placebo control reveals. The placebo is the most consequential limitation we report. Our random-direction baselines establish that the bucket-level effect is not a generic response to large perturbations, but they cannot establish that the pin direction uniquely encodes pin-related output behavior. Our placebo shows that other meaningful directions exist, namely a nearly-orthogonal line-attack direction produces a larger bucket-level effect than pin at matched magnitude in 96% of bootstrap resamples. Pin and line-attack information are therefore encoded in nearly-disjoint directions despite line-attack being the superset of pin as a tactical concept, demonstrating that the trained residual stream contains distinct linear representations for distinct board-geometry concepts. This fits broader work showing that high-dimensional residual streams support many simultaneous task-relevant features (Bricken et al., 2023; Templeton et al., 2024). Sparse-autoencoder analysis at the relevant layer would clarify how many independent king-geometry directions exist and whether pin and line-attack separate cleanly into distinct features.

From bucket flips to move recommendations. The per-action result (Section 4) reveals a related but distinct issue: pin-direction patching reshapes the model's value distribution within the implicit move ranking but rarely enough to flip which legal move is on top (2.8% vs. the 36.8% bucket-flip rate). Two readings are consistent: the searchless model may use pin information primarily for fine-grained value calibration rather than for binary move selection (the latter typically dominated by other features); or our matched-pair test positions may often have one strongly-best legal move whose value advantage exceeds the perturbation produced by patching, so the 2.8% rate reflects positions where the top two moves are within the pin-direction's effective range. The random-direction baseline at the per-action level sharpens this picture: pin's $1.73\times$ multiplier over random is much smaller than the $6.2\times$ multiplier at the bucket level. The attenuation factor of $\sim 3.6\times$ from bucket to move is itself a quantitative measurement of how directional-specificity findings shrink as the metric moves closer to behavior.

6.2. Operational caveats

We study absolute pins only, as relative pins (where moving the pinned piece is legal but loses material) require symbolic reasoning beyond `python-chess` and we leave this to future work. The underlying searchless transformer checkpoints (Ruoss et al., 2024) are publicly available.

7. Conclusion

The 270M searchless chess transformer linearly represents absolute pins beyond what is recoverable from raw board geometry, and patching a layer-6 pin direction produces a monotonic, layer-localized return-bucket effect. Stronger controls narrow the interpretation: a semantically related line-attack direction produces larger bucket flips than pin in 96% of bootstrap resamples, recommended moves change on only 2.8% of positions despite the 36.8% bucket-flip rate, and probe and patching peaks occur at different layers in both 270M and 9M.

These findings expand Hewitt & Liang's methodological caution: probe accuracy does not localize causal computation; directional specificity against random baselines does not establish concept-specific encoding; and return-bucket flipping does not establish recommended-move change. Each type of evidences ends up capturing something narrower than the strong reading suggests. Taking all three into account, and doing so cautiously, is necessary for future takeaways. Future work should test whether SAE or DAS analyses separate pin and line-attack features, and whether the same probe–patching and bucket-vs-move gaps appear for other tactical motifs and chess models.

Impact Statement

This paper presents work whose goal is to advance the field of mechanistic interpretability. Our analysis is conducted on a publicly available chess model and uses publicly available game data; we do not foresee specific ethical or societal consequences beyond those generally associated with interpretability research, namely the potential for improved trustworthiness and debuggability of larger neural systems.

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A. Full Layer-Wise Tables (270M Model)

Table 5. Layer-wise probe results on the 270M model. AUC is test-set ROC AUC; CI is 95% bootstrap. Random AUC is the AUC of a probe trained on shuffled labels. Selectivity is the gap between true and random AUC. Bold: peak.

Layer	AUC	95% CI	Random	Selectivity
0	0.634	[0.619, 0.648]	0.517	0.085
1	0.883	[0.874, 0.891]	0.497	0.305
2	0.911	[0.904, 0.918]	0.524	0.314
3	**0.919**	[0.912, 0.925]	0.511	**0.328**
4	0.918	[0.911, 0.925]	0.510	0.331
5	0.911	[0.903, 0.918]	0.502	0.319
6	0.880	[0.871, 0.888]	0.498	0.305
7	0.853	[0.843, 0.862]	0.535	0.247
8	0.832	[0.821, 0.841]	0.503	0.246
9	0.817	[0.806, 0.827]	0.508	0.235
10	0.802	[0.792, 0.814]	0.526	0.207
11	0.791	[0.780, 0.802]	0.507	0.217
12	0.781	[0.770, 0.793]	0.499	0.207
13	0.776	[0.764, 0.787]	0.496	0.201
14	0.769	[0.757, 0.781]	0.515	0.192
15	0.763	[0.751, 0.775]	0.498	0.194
16	0.759	[0.746, 0.771]	0.517	0.180

Table 6. Layer-wise activation patching on the 270M model, evaluated on 500 held-out pin→no-pin pairs. KL: KL divergence between patched and base return-bucket distributions. Flip rate: fraction of positions where the argmax bucket of the model's value-prediction output changes; CI is Wilson 95% binomial. Bold: peak flip rate.

Layer	Mean KL	Flip rate	95% CI	Logit diff
0	0.146	0.140	[0.112, 0.173]	0.014
1	0.196	0.138	[0.111, 0.171]	0.016
2	0.155	0.134	[0.107, 0.167]	0.017
3	0.196	0.168	[0.138, 0.203]	0.015
4	0.029	0.176	[0.145, 0.212]	0.003
5	0.029	0.284	[0.246, 0.325]	0.000
6	0.042	**0.368**	[0.327, 0.411]	0.002
7	0.040	0.308	[0.269, 0.350]	0.002
8	0.005	0.114	[0.089, 0.145]	0.006
9	0.003	0.092	[0.070, 0.121]	0.009
10	0.002	0.066	[0.047, 0.091]	0.011
11	0.001	0.048	[0.032, 0.070]	0.014
12	0.002	0.064	[0.046, 0.089]	0.017
13	0.002	0.066	[0.047, 0.091]	0.017
14	0.002	0.058	[0.041, 0.082]	0.016
15	0.002	0.060	[0.042, 0.084]	0.015
16	0.002	0.078	[0.058, 0.105]	0.015

B. 9M Model Results

For completeness, we report the analogous probing and patching tables for the 9M variant of the architecture. The qualitative pattern matches the 270M results: an inverted-U probing curve, a probe peak in the early-mid layers, and a patching peak two layers downstream of the probe peak. Absolute numbers are weaker: peak AUC 0.825 vs. 0.919, peak flip rate 0.306 vs. 0.368. The qualitative ordering is similar: in both models, the patching peak occurs downstream of the probe peak. However, the fractional depths differ substantially: the 9M probe and patching peaks occur at layers 5/8 ($\approx$0.63) and 7/8 ($\approx$0.88), while the 270M peaks occur at layers 3/16 ($\approx$0.19) and 6/16 ($\approx$0.38). The 270M model shows peak pin decodability earlier in fractional depth, consistent with the observation that larger models form abstract concepts in earlier (relative) layers (Templeton et al., 2024).

The 9M logit-diff column shows small negative values at most layers (magnitudes $|0.001–0.006|$), in contrast to the 270M model where logit diffs are slightly positive at the patching plateau. Both magnitudes are within bootstrap noise of zero

Table 7. Layer-wise probe results on the 9M model.

Layer	AUC	95% CI	Random	Selectivity
0	0.634	[0.619, 0.648]	0.517	0.085
1	0.776	[0.763, 0.787]	0.485	0.219
2	0.775	[0.763, 0.787]	0.499	0.205
3	0.798	[0.787, 0.808]	0.502	0.232
4	0.811	[0.800, 0.820]	0.523	0.214
5	**0.825**	[0.814, 0.835]	0.490	**0.253**
6	0.806	[0.795, 0.817]	0.520	0.217
7	0.756	[0.743, 0.768]	0.511	0.186
8	0.722	[0.709, 0.735]	0.513	0.158

Table 8. Layer-wise patching results on the 9M model (500 held-out pairs).

Layer	Mean KL	Flip rate	Logit diff
0	0.017	0.096	-0.002
1	0.022	0.110	-0.004
2	0.007	0.118	-0.005
3	0.019	0.162	-0.005
4	0.021	0.182	-0.004
5	0.018	0.248	-0.001
6	0.015	0.214	-0.006
7	0.023	**0.306**	-0.001
8	0.004	0.128	0.015

($| < 0.01|$ on bootstrap CIs from the same protocol used for the 270M numbers), and we read this as essentially-zero logit-diff in both models, consistent with the interpretation in the main text (Section 4). We do not take the sign difference between models to be meaningful given these magnitudes.

Table 9. Dose–response sweep on the 9M model at the patching peak (layer 7), 500 held-out pairs.

α	Flip rate	Mean KL
0.0	0.000	0.000
0.5	0.162	0.006
1.0	0.316	0.023
2.0	0.490	0.094
5.0	0.752	0.702

Table 10. Cross-layer + norm-controlled patching on the 9M model between probe peak (layer 5) and patching peak (layer 7), 500 held-out pairs. $\|\Delta\mathbf{h}_5\| = 85.10$, $\|\Delta\mathbf{h}_7\| = 775.00$ (ratio ≈ 9.1).

Condition	Flip rate	Mean KL
$\Delta\mathbf{h}_5$ native @ L5 (own)	0.236	0.019
$\Delta\mathbf{h}_7$ native @ L7 (own)	**0.316**	0.023
$\Delta\mathbf{h}_5$ native @ L7 (cross)	0.020	0.000
$\Delta\mathbf{h}_7$ native @ L5 (cross)	0.618	0.236
$\Delta\mathbf{h}_5$ rescaled to $\|\Delta\mathbf{h}_7\|$ @ L7	0.232	0.016
$\Delta\mathbf{h}_7$ rescaled to $\|\Delta\mathbf{h}_5\|$ @ L5	0.120	0.004

The 9M cross-layer pattern matches the 270M's qualitatively: same-layer patching dominates, native-norm cross-layer patching is asymmetric (magnitude-dominated), and norm-matched cross-layer patching falls in between. Quantitatively, the 9M norm-matched cross-layer effect (0.232 with $\Delta\mathbf{h}_5 \to L7$) is closer to the own-layer effect (0.316 at L7) than in the 270M (0.100 cross-matched vs. 0.368 own at L6), suggesting that the layer-specific specialization of pin encodings strengthens with model scale; we caution that two model sizes is weak evidence for this trend.

C. Probe Robustness Controls

To validate that probe accuracy reflects pin-relevant linear structure rather than probe overfitting, model architecture, or input encoding, we report three control experiments: an MLP-probe upper bound, an untrained-model control, and a bitboard-baseline probe.

MLP probe upper bound. Table 11 compares linear-probe AUC at each model's peak layer to a 1-hidden-layer MLP probe (hidden size selected from $\{64, 128, 256\}$ on validation AUC, ℓ_2 weight decay 10^{-3}). For comparison we also include the bitboard baseline described in Section 3.3, which trains the same probes directly on raw piece-on-square indicator features. The MLP gain over linear probing is small at both transformer scales, indicating that pin information at the model's peak layer is essentially linearly accessible.

Table 11. Linear vs. MLP probe comparison at each model's peak layer, plus the bitboard baseline (probes trained directly on raw piece-on-square features). CI: 95% bootstrap confidence interval.

Source	Linear AUC	MLP AUC	Gap
270M, layer 3	0.919 [0.912, 0.925]	0.926 [0.920, 0.932]	+0.007
9M, layer 5	0.825 [0.814, 0.835]	0.847 [0.837, 0.856]	+0.022
Bitboard baseline	0.774 [0.763, 0.786]	0.830 [0.819, 0.840]	+0.056

Untrained-model control. Tables 12 and 13 report probe AUC by layer on randomly-initialized 9M and 270M models (no checkpoints loaded; same architectures, same FEN tokenization, same probing pipeline). The peak AUCs of 0.737 (9M, layer 1) and 0.776 (270M, layer 1) are both well above chance, indicating that the FEN tokenization plus random attention is sufficient to make pin labels partially linearly recoverable. Training adds +0.088 AUC over this baseline at the 9M scale and +0.143 AUC at the 270M scale; we treat these gaps as the training-attributable component of the trained models' linear pin representations.

Table 12. Layer-wise probe results on a randomly-initialized 9M model. The peak at layer 1 reflects the architectural + tokenization prior; subsequent layers do not improve over this prior in the absence of training. Bold: peak.

Layer	AUC	95% CI	Random AUC	Selectivity
0	0.634	[0.619, 0.648]	0.516	0.086
1	**0.737**	[0.725, 0.750]	0.509	**0.171**
2	0.732	[0.720, 0.744]	0.527	0.153
3	0.732	[0.720, 0.745]	0.507	0.164
4	0.732	[0.719, 0.744]	0.527	0.157
5	0.727	[0.715, 0.741]	0.491	0.172
6	0.729	[0.716, 0.741]	0.529	0.145
7	0.720	[0.708, 0.734]	0.460	0.198
8	0.719	[0.707, 0.732]	0.497	0.159

D. Dataset Statistics

The balanced probing dataset contains 38,312 positions split 70/15/15 by `game_id` hash. The pin rate per split is 50.16% (train), 49.28% (val), 49.99% (test). Material counts and move numbers were stratified into 4 buckets each, yielding 16 strata; within each stratum we sampled equal numbers of pin-present and pin-absent positions.

The paired dataset contains 2,000 paired positions from 368 unique games. Mean material counts: 21.1 (pin) vs. 21.0 (no-pin). Mean material difference between paired positions: 0.34 pieces. Move-number range: 11–204, mean 52.1.

E. Hyperparameters

Probe training. L2-regularized logistic regression (scikit-learn), L-BFGS solver, $C \in \{0.01, 0.1, 1.0, 10.0\}$, selected on val AUC. `max_iter`=1000. Activations are mean-pooled across the 79 token positions, stored in float16, then converted to float32 for training. No further normalization.

Activation patching. Mean direction $\Delta\mathbf{h}_\ell$ estimated from 1,500 train pairs and evaluated on 500 held-out pairs. Intervention

Table 13. Layer-wise probe results on a randomly-initialized 270M model. The peak at layer 1 again reflects the architectural + tokenization prior; later layers do not improve over this prior. Bold: peak.

Layer	AUC	95% CI
0	0.634	[0.619, 0.648]
1	**0.776**	[0.764, 0.788]
2	0.775	[0.764, 0.787]
3	0.775	[0.764, 0.786]
4	0.773	[0.762, 0.784]
5	0.775	[0.764, 0.786]
6	0.774	[0.763, 0.785]
7	0.773	[0.762, 0.784]
8	0.770	[0.759, 0.782]
9	0.771	[0.760, 0.783]
10	0.770	[0.759, 0.782]
11	0.771	[0.761, 0.783]
12	0.771	[0.760, 0.783]
13	0.771	[0.761, 0.783]
14	0.771	[0.761, 0.782]
15	0.770	[0.759, 0.782]
16	0.769	[0.758, 0.780]

is single-layer (no multi-layer composition). The headline flip rate uses coefficient $\alpha = 1$, corresponding to the natural "move from pin manifold to no-pin manifold" translation in residual space; we additionally report a coefficient sweep ($\alpha \in \{0, 0.5, 1, 2, 5\}$) at layer 6 for dose–response analysis (Section 4).

Bootstrap CIs. 1,000 resamples of the test set with replacement; report 2.5/97.5 percentiles.